

Google OAuth Setup — Step by Step

How the Google Cloud project for SparkClone was created · Account: kbalaji9738@gmail.com · August 11, 2026

This guide records exactly how SparkClone's Google integration was configured, so you can repeat it for any future project (or another Google account). Everything below is **free** — no billing account or credit card is required for the Gmail and Drive APIs at personal usage levels.

What you end up with: an OAuth **Client ID** and **Client Secret** — a pair identifying your app to Google so it can show the "Allow this app to access your Gmail/Drive?" consent screen. This is *not* the same as an API key: no password or key of yours is ever given to the app; Google issues limited-power tokens after you approve.

Part 1 — Create the project

STEP 1 — OPEN THE PROJECT CREATION PAGE

Go to <https://console.cloud.google.com/projectcreate> and make sure the account menu (top-right avatar) shows the Google account you want the agent to access — here, **kbalaji9738@gmail.com**. If another account is active, click the avatar and switch accounts first.

STEP 2 — NAME AND CREATE IT

- Project name: `sparkclone`
- Parent resource: *No organization* (default for personal Gmail accounts)
- Click **Create**, wait a few seconds, then pick the new project from the project dropdown (top-left). Result: project ID `sparkclone-505213` .

Personal accounts have a default quota of ~25 projects. This account had 1 slot remaining, which this project used — delete unused projects at *IAM & Admin* → *Manage Resources* if you ever need another.

Part 2 — Enable the APIs

STEP 3 — GMAIL API

Menu → **APIs & Services** → **Library**, search "Gmail API", open it, click **Enable**. (Direct link: console.cloud.google.com/apis/library/gmail.googleapis.com)

STEP 4 — GOOGLE DRIVE API

Same library, search "Google Drive API", click **Enable**.
(`console.cloud.google.com/apis/library/drive.googleapis.com`)

Part 3 — Configure the OAuth consent screen

STEP 5 — START THE WIZARD

Menu → **APIs & Services** → **OAuth consent screen** (Google now calls this "Google Auth Platform"). Click **Get started**. Fill the four sections:

Section	Value used
App Information	App name <code>SparkClone</code> , support email <code>kbalaji9738@gmail.com</code>
Audience	External (personal Gmail accounts can't use Internal)
Contact Information	<code>kbalaji9738@gmail.com</code>
Finish	Tick "I agree to the Google API Services: User Data Policy", Continue, Create

STEP 6 — ADD YOURSELF AS A TEST USER

Go to the **Audience** page (left sidebar). Under **Test users** click **+ Add users**, enter `kbalaji9738@gmail.com` , and **Save**. The app stays in **Testing** publishing status — only listed test users can sign in, and no Google verification review is needed.

The one Testing-mode quirk: for External apps in Testing, Google expires refresh tokens after about **7 days**. Practically: you'll click "Sign in with Google" in SparkClone again roughly once a week. Acceptable for a personal self-hosted tool; the alternative (publishing to Production) triggers a Google review because Gmail scopes are "restricted", which isn't worth it here.

Part 4 — Create the OAuth client

STEP 7 — CREATE THE CLIENT ID

Sidebar → **Clients** → + **Create client**:

- Application type: **Web application**
- Name: `sparkclone-web`
- Authorized redirect URIs → + **Add URI**:
`http://localhost:8000/auth/google/callback`
(this is the FastAPI route Google sends the user back to after consent — it must match the backend exactly, including port and path)
- Authorized JavaScript origins: left empty (not needed for a server-side flow)
- Click **Create**.

STEP 8 — COPY THE CREDENTIALS IMMEDIATELY

A dialog shows the **Client ID** and **Client Secret**. The secret is shown only once — copy both (or Download JSON) before clicking OK. If you lose the secret, open the client on the Clients page and use *Reset secret*.

Client ID	<code>145271900279-gq1dvut9hkc21kkjk873v68u766jg1ld.apps.googleusercontent.com</code>
Client secret	<code>GOCSPX-*****</code> (stored in the project's <code>.env</code> ; not printed here since PDFs get shared)

Part 5 — Wire it into SparkClone

STEP 9 — PUT THE VALUES IN `.ENV`

```
GOOGLE_CLIENT_ID=145271900279-gq1dvut9hkc21kkjk873v68u766jg1ld.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=<the GOCSPX-... secret>
OAUTH_REDIRECT_URL=http://localhost:8000/auth/google/callback

# random values (generated with: openssl rand -hex 24)
SPARK_SECRET_KEY=<48 hex chars> # signs session cookies + encrypts stored requests
SPARK_API_TOKEN=<48 hex chars> # bearer-token fallback for scripts/API clients
```

STEP 10 — THE IMAP/SMTP BLOCK: LEAVE IT EMPTY

The `.env` still contains `IMAP_HOST` / `SMTP_HOST` / `...` fields. These power the **old** email path (app-password based). With Google OAuth, the Gmail API replaces them, so they stay blank as an optional fallback. `NOTIFY_EMAIL` is kept filled so notifications know their destination. Until SMTP is configured or Gmail sending is built, the `notify` tool logs to the server console instead of emailing.

Result

Item	Status
Google Cloud project <code>sparkclone</code> (<code>sparkclone-505213</code>)	Created
Gmail API + Google Drive API	Enabled
OAuth consent screen "SparkClone" (External, Testing)	Configured
Test user kbalaji9738@gmail.com	Added
OAuth client <code>sparkclone-web</code> + localhost redirect URI	Created
Credentials saved to project <code>.env</code>	Done
Cost	₹0 / \$0 — no billing account needed